

HYPERTENSION

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INTRODUCTION

Importance

Hypertension or high blood pressure is a chronic medical condition in which the arterial blood pressure is elevated. Persistent hypertension is one of the key risk factors for cardiovascular diseases such as heart attack, stroke, and heart failure as well as other diseases like kidney failure. It is often known as a silent killer as it rarely causes symptoms, and many people go undiagnosed. Dietary and lifestyle changes can improve blood pressure control and decrease the risk of associated health complications, although drug treatment may be necessary in patients for whom lifestyle changes prove ineffective or insufficient (WHO, 2013).

Ischaemic heart disease (IHD) and stroke are the third and fourth leading causes of death in Singapore in 2022. In addition, hypertensive diseases (including hypertensive heart disease) together constitute the fifth leading cause of death.

Epidemiology

Hypertension is a highly prevalent risk factor for the development of cardiovascular disease (CVD), affecting an estimated 1.28 billion people worldwide.⁴ In Singapore, the 2022 National Population Health Survey reported that over one in three residents (37%) aged 18 to 74 years had hypertension, and that more than half of this group (53%) were previously undiagnosed.

The crude prevalence of hypertension showed a significant increase from 2010 to 2021-2022 while the upward trend for the age-standardised prevalence was not significant in the same period (crude: 19.8% in 2010, 37.0% in 2021-2022; age-standardised: 19.8% in 2010, 32.4% in 2021-2022). More males (44.0%) were hypertensives compared with females (30.2%) during the period 2021-2022. Prevalence of hypertension increased with age; starting at around 8.1% for those aged 18 to 29 years to 76.8% among those aged 70 to 74 years. Among all residents with hypertension, more than half (53.5%) of them had not been previously diagnosed with hypertension. Among the known hypertensives who attended health examination, about two-thirds (64.8%) had poor control of their blood pressure.

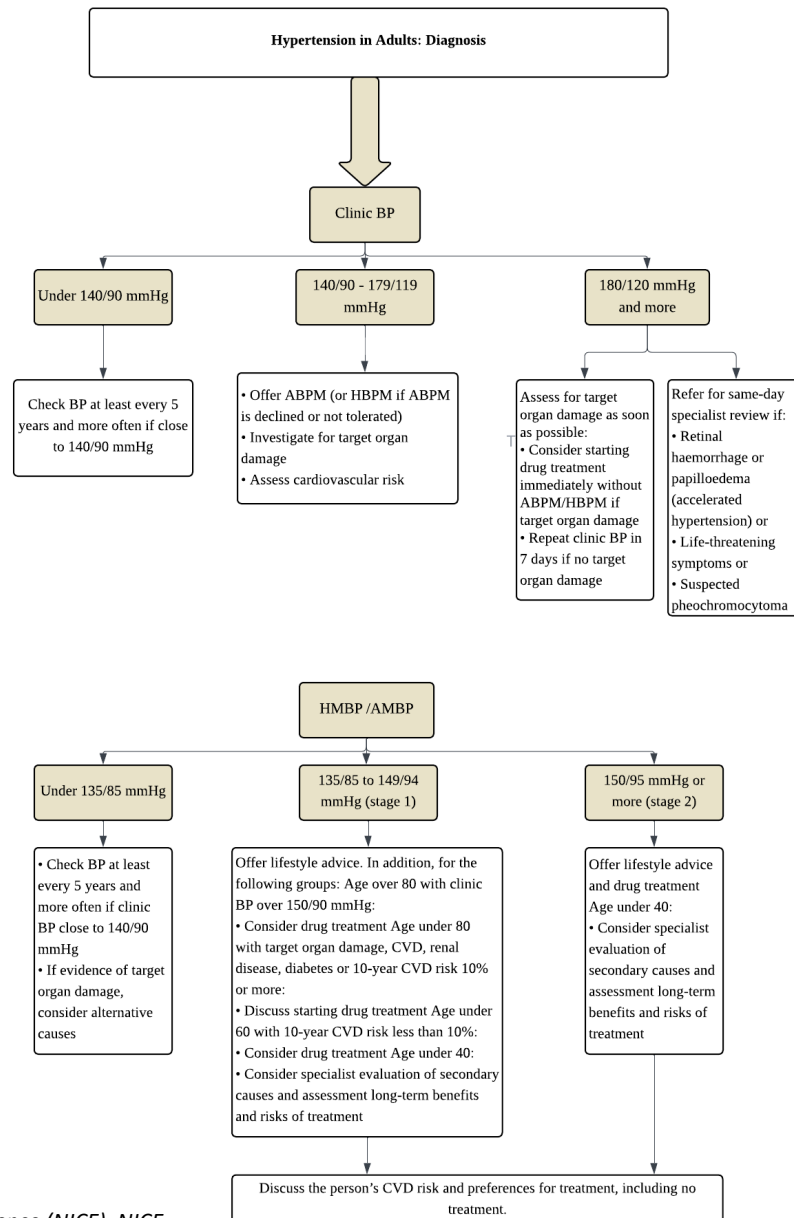
SCREENING AND DIAGNOSIS

Recommended Screening

Any patient aged ≥ 18 years during any clinical visit.

Initial Screening Blood Pressure (BP) (Average of 2 or More Readings)

Systolic BP (mmHg)	Diastolic BP (mmHg)	Category	Recommended Action
< 130	< 85	Normal	"Normal BP". Advise BP check biennially.
130 - 139	85 - 89	High-Normal BP	Advise lifestyle modification. Check BP annually or more frequently if cardiovascular risk factors are present.
140 - 159	90 - 99	Grade 1 Hypertension	For those without cardiovascular risk factors, try lifestyle modification first for 3 - 6 months (shorter for those with cardiovascular risk factors). For those at high risk with established cardiovascular or renal disease, diabetes mellitus or target organ damage, initiate drug treatment with lifestyle measures at the same time.
160 - 179	100 - 109	Grade 2 Hypertension	For those at low risk with 0 to 2 cardiovascular risk factors, can try lifestyle modification first for several weeks; otherwise, initiate drug treatment with lifestyle measures at the same time.
≥ 180	≥ 110	Grade 3 Hypertension	Initiate drug treatment with lifestyle measures at the same time.
≥ 140	< 90	Isolated Systolic Hypertension	Isolated systolic hypertension is graded according to the same ranges of systolic BP as the above, and the corresponding recommendations apply.



Source: National Institute for Health and Care Excellence (NICE). NICE Guideline 136 on Hypertension in adults: Diagnosis and management

Blood Pressure Measurement

Blood pressure should be measured at rest several times on several occasions, with the patient in the supine or sitting position, using a non-invasive manometer. Accuracy of non-mercury devices should be ensured through periodic calibration by the vendor.

The following procedures should be observed when recording the blood pressure during the initial assessment:

- Measure BP in both arms; all subsequent readings to be taken on the arm with the higher reading.
- Measure BP in sitting (or supine) position, and 2 minutes after standing, for elderly and diabetic patients.

1. Preparation and Positioning for BP Measurement

- Patient should rest 5 minutes before measuring the blood pressure. Empty bladder if needed.
- Patient to refrain from smoking, ingesting caffeine, physical exercise or eating at least 30 minutes before the measurement.
- Patient to sit comfortably in a quiet room, with back supported, and both feet on the floor with legs uncrossed.
- Select proper BP cuff size: BP cuff should be long enough to encircle the circumference of the patient's arm completely and width of BP cuff should cover two-third of the length of patient's arm.
- Support forearm with palm facing upwards. Place cuff over exposed arm and 2.5cm above the antecubital fossa (elbow crease). Cuff at the level of the heart. Wrap and secure the cuff snugly around the arm. Ensure the tubing from the cuff is not obstructed or kinked.

2. Office Blood Pressure Measurement using Auscultatory Method

- Patient should be advised not to talk or move during BP measurement.
- The cuff should be inflated to a pressure approximating 30 mmHg greater than the systolic reading and is indicated by the disappearance of the brachial pulse during palpation.
- The stethoscope should be placed lightly over the brachial artery.
- The cuff should be deflated at the rate of 2-3 mmHg per heartbeat.
- The diastolic reading corresponds to Korotkoff phase V.

3. Office Blood Pressure Measurement using Electronic BP Machine

- Patient should be advised not to talk or move during BP measurement.
- Press the "Start" button and the unit automatically inflates the cuff to generally 20 mmHg above patient's systolic pressure.
- The cuff will gradually deflate automatically as the sensor detects the patient's systolic and diastolic BP.
- When the measurement is complete, the blood pressure and pulse rate will be displayed in the LCD screen.
- If first BP reading is abnormal, take two additional BP readings with at least 1 minute between them and calculate the average of the last 2 measurements.

Office blood pressure measurement



1
Measure after 5 min seated comfortably in a quiet environment



2
Use a validated device with an appropriate cuff size based on arm circumference



3
Place the BP cuff at the level of the heart with the patient's back and arm supported



8
Assess for orthostatic hypotension at 1st visit and thereafter by symptoms



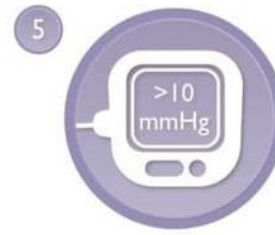
4
Measure BP three times (1–2 min apart) and average the last 2 readings



7
Record heart rate and exclude arrhythmia by pulse palpation



6
Measure BP in both arms at the 1st visit to detect between arm differences



5
Obtain further measurements if the readings differ by >10 mmHg

Initial Assessment – History, Examination and Investigations

Clinical Assessment to determine if:

1. There is a secondary cause for the hypertension.
2. There is target organ damage (TOD).
3. There are other cardiovascular risk factors present.
4. In the absence of conditions known to influence cardiovascular risk, use SG-FRS-2023 to calculate patient's 10 year risk.

	OTHER CV RISK FACTORS	SECONDARY CAUSES OF HYPERTENSION	TARGET ORGAN DAMAGE
HISTORY	Age - Male > 45 yr - Female > 55 yr Smoking Personal history of - Diabetes Mellitus - Hyperlipidaemia - TC > 6.2 mmol/L - LDL-C > 4.1 mmol/L - HDL-C < 1.0 mmol/L Family history of: - Premature IHD in 1st degree relative - Male < 55 yr - Female < 65 yr	Age of onset < 30 yr Poor response to generally effective therapy Personal history of - Renal disease Family history of - Renal disease Drug history - Alcohol, NSAIDs, oral contraceptives / steroids Symptoms suggestive of phaeochromocytoma (4Hs) - Hypertension (labile or sustained), heart palpitations, headaches, hyperhidrosis	Personal history of - Ischaemic heart disease - Stroke / transient ischaemic attack (TIA) Symptoms suggestive of - Congestive cardiac failure - Stroke / TIA - Intermittent claudication
EXAMINATION	Height and Weight - BMI ≥ 27.5 BP - Grade 1 - 3	BP (Grade 2 or higher) Fundoscopy (Keith-Wagener Grade 2 or higher) Enlarged kidneys Renal bruit Radio-femoral delay Signs suggestive of Cushing's, thyroid disease, neurofibromatosis	Abdominal aortic aneurysm Peripheral pulses Signs suggestive of - Heart failure - Carotid bruit - Fundi - Neurological deficit
INVESTIGATIONS [Hypertension Panel + Electrocardiogram (ECG)]	Lipid levels Blood glucose level Serum creatinine / electrolytes	ECG for Left Ventricular Hypertrophy (LVH) Serum creatinine / serum potassium (hypokalaemia) Urine dipstick test for blood and protein (if positive, follow-up with urine microscopy / urine protein-creatinine ratio respectively)	ECG (for IHD and LVH) Serum creatinine, electrolytes Urine dipstick test (if positive, to follow-up with urine microscopy / urine protein-creatinine ratio)

- The more important items have been indicated in **bold** and should be done in **all patients**.
- It is not necessary to do all the items in one consultation - they can be covered over several consultations. A full history should be taken once the screening BP is more than 130/80 mmHg.

MANAGEMENT

Goals of Treatment

- Prevent cardiovascular and renal complications.
- Treat the whole patient and associated conditions / risk factors.
- Avoid drug side effects.
- Majority may require two or more medications to reach the goal.
- Aim for BP control within 3 months.

Treatment Targets

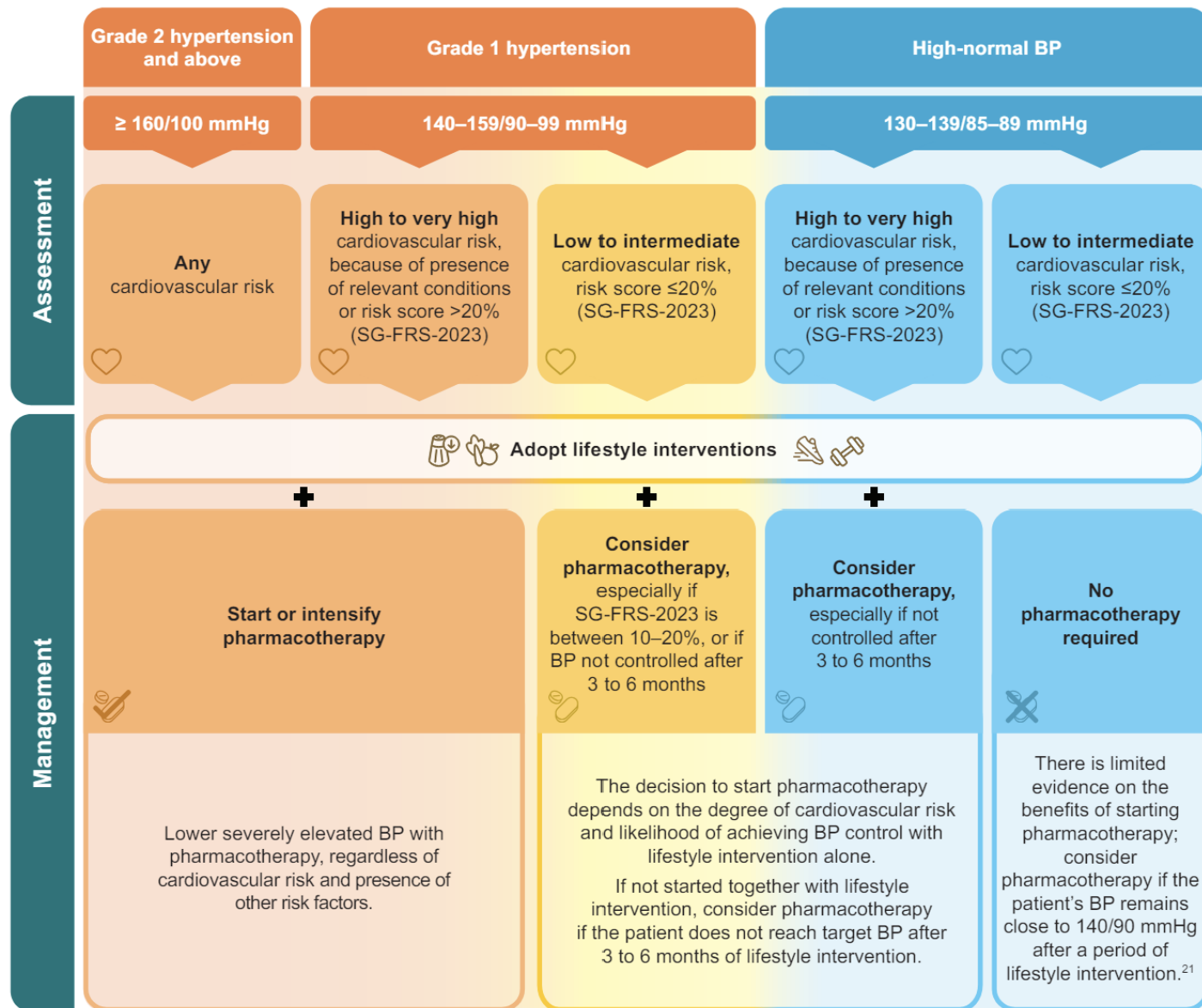
Patient stratification based on comprehensive assessment	BP targets (clinic reading)
High to very high cardiovascular risk • Patients with CVD, CKD, DM, or HMOD OR risk score >20% (SG-FRS-2023)	<130/80 mmHg
Low to intermediate cardiovascular risk • Patients with none of the above risk factors/comorbidities OR risk score ≤20% (SG-FRS-2023)	<140/90 mmHg Consider going lower as tolerated (e.g., <130/80 mmHg)
Older age (e.g., >80 years old), frail, orthostatic hypotension, or limited life expectancy	Consider less stringent BP targets (e.g., <150/90 mmHg)

Source: ACE Clinical Guidance (published 15 December 2023)

Special conditions:

- < 150/100 mmHg in pregnant patients without target organ damage (do not decrease diastolic BP to < 80 mmHg).
- < 140/90 mmHg in pregnant patients with target organ damage.
- < 220/120 mmHg during 1st 24 hrs of acute stroke (lower with care by 10 - 15%).
- Lower by 10/5 mmHg if BP > 140/90 mmHg after acute phase of stroke.

Initiation of Pharmacotherapy



Reference for Corresponding Values of Clinic and Non-Clinic BP Readings

Clinic (mmHg)	HMBP or Daytime AMBP (mmHg)	Night-time AMBP (mmHg)	24-hour AMBP (mmHg)
120/80	120/80	100/65	115/75
130/80	130/80	110/65	125/75
140/90	135/85	120/70	130/80
160/100	145/90	140/85	145/90

HMBP: Home Blood Pressure, AMBP: Ambulatory Blood Pressure

The above table reflects the average clinic and non-clinic BP values across untreated and treated individuals from non-Asian and Asian ethnicities.

BP readings taken in non-clinic settings tend to be lower than readings taken in clinic. The difference between clinic and non-clinic BP readings is not fixed and decreases as clinic BP decreases. The above table should be interpreted with caution as the information reflects the average clinic and non-clinic BP values across untreated and treated individuals from non-Asian and Asian ethnicities.

Recommended Follow-Up Interval

Criteria	Recommended TCU Frequency	Recommended for alternate Dr / CM visit
• Good blood pressure control, AND • Does not have complication	6 months	√
• Good blood pressure, AND • Elderly or has complications such as IHD, CVA, renal impairment	3 - 4 months	√
• Adherent to follow-up and treatment, AND • With or without co-morbidities / complications, AND • Stable but sub-optimal control related to individual targets for BP, HbA1c and Cholesterol over the past 4 -6 months	3 - 4 months	√
• ACEi / ARB initiation or up-titration (K and Cr to be done in 2 weeks)	2 - 4 weeks	√
• Poor blood pressure control, AND • Requires titration of medication	2 – 4 weeks	

Lifestyle Modification and Patient Education

Lifestyle modification is an important component in the management of hypertension and should be recommended to the patients. Wherever possible, use a team-based approach to manage a patient with hypertension, involving trained nurses and pharmacists with medical practitioners. (Grade A, Level 1+, 1)

All newly diagnosed hypertensive patients should be referred to a care manager for education regarding their condition. Consider referral to a dietitian for dietary approaches, e.g., DASH diet.

If appropriate, tell the patient:

"You should stop smoking."

"You should lose weight."

"You should exercise regularly."

"You should not add salt to your food."

Important health education topics include:

- target blood pressure
- benefits and side effects of treatment
- risks of hypertension
- the importance of long-term adherence to drug treatment (if any)
- stress reduction

Non-Pharmacological Treatment

- Recommend lifestyle changes to all hypertensive patients, and in patients with high normal BP. However, drug treatment should not be delayed without reason beyond 3 to 6 months if indicated. (Grade A, Level 1+, 1)
- Advise patient to restrict salt intake to 5 to 6 g per day. (Grade A, Level 1+, 1)
- Moderate alcohol consumption to no more than 2 standard drinks per day for men, and to no more than 1 standard drink per day for women. (Grade A, Level 1+, 1)
- Increase the consumption of vegetables, fruits, low-fat dairy products, and decrease the intake of saturated and total fats. (Grade A, Level 1+, 1)
- Unless contraindicated, advise patients to reduce weight to a BMI below 23 kg/m² and to a waist circumference below 90 cm in men, and below 80 cm in women (for Asians). (Grade B, Level 2+, 1)
- Advise patients to do at least 30 minutes of moderate dynamic exercise 5 to 7 days per week. Any physical exercise above the basal level, up to about 150 minutes a week, confers incremental cardiovascular and metabolic benefits, including BP reduction. (Grade A, Level 1+, 1)
- Advise and offer assistance to all smokers to quit smoking. (Grade A, Level 1+, 1)

Pharmacological Treatment

The choice of the drug used to treat hypertension depends on the patient's pre-existing medical conditions (if any), the associated compelling indications and contraindications, e.g., avoid beta blockers for asthmatics.

INITIAL DRUG CHOICES

Uncomplicated Hypertension

- ACE Inhibitor (ACE-I) / Angiotensin Receptor Blocker (ARB)
- Calcium Channel Blocker (CCB)
- Diuretic

*Compelling Indications

Diabetes Mellitus

- ACE-I / ARB

Chronic Kidney Disease / Proteinuria

- ACE-I / ARB

Heart Failure

- ACE-I / ARB (preferred first line)
- Beta-blocker (preferred first line)
- Diuretic

Isolated systolic hypertension (older persons)

- Diuretic
- Long-acting CCB

Myocardial infarction

- Beta-blocker
- ACE-I / ARB (LV dysfunction)

- Start with a low-dose long-acting once-daily drug and titrate dose.
- Low-dose combinations may be appropriate.

Contraindications

Asthma / Bronchospasm

- Beta-blocker (consider cardio-selective beta-blockers if indicated and use with caution)

2° or 3° Heart Block

- Beta-blocker
- Verapamil

Gout

- Diuretic

Bilateral Renal Artery Stenosis

- ACE-I, ARB

Pregnancy

- ACE-I, ARB, Diuretic

Initial Drug Choices

Not at desired blood pressure

No response or
troublesome side-effects

Substitute another drug from
different class

Inadequate response
But well tolerated

Add a second agent from a
different class
(diuretic if not already used)

Not at desired blood pressure

- Add one or more agents from other classes
- Consider referral to a hypertension specialist

After any drug is started or its dosage increased, always ask the patient if there are any adverse effects from the drug.

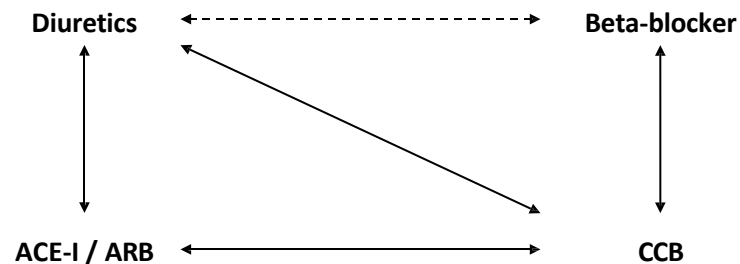
After starting the following drugs, monitor:

ACE inhibitor/ARB: Serum creatinine and potassium levels after 2 - 4 weeks

Diuretic: Serum potassium and sodium levels (check serum creatinine in advanced CKD patients)

**Refer to the respective CPGs for further guidance*

Drug Combinations:



- Consider initiating low-dose dual therapy from two different anti-hypertensive medication classes based on required BP reduction and cardiovascular risk.
- Beta-blocker + ACE-I / ARB does not produce synergistic BP reduction.
- ACE-I + ARB **worsens** GFR **and potentiates hyperkalaemia** in patients and **the combination of both drugs** should be avoided.
- Beta-blocker + diuretic increases risk of developing diabetes mellitus.

When to begin drug treatment?

- For high or very high-risk individuals, begin immediate drug treatment for hypertension when other risk factors or conditions are present. (Grade A, Level 1++, 1)
- For medium risk individuals, monitor the BP and other risk factors for several weeks, and obtain further information, before deciding whether to begin drug treatment. (Grade B, Level 1+, 1)
- For low-risk individuals, observe the patient over a significant period of time before deciding whether or not to begin drug treatment. (Grade B, Level 1+, 1)

How to begin drug treatment?

- Begin appropriate combination treatment in patients whose pre-treatment BP is raised (i.e., $\geq 140/90$ mmHg), and specifically in patients whose BP is severely raised ($\geq 180/110$ mmHg), as they will require two or more drugs for adequate BP control. (Grade B, Level 2+, 1)
- Initiate treatment at low doses, either singly or as a two-drug combination, to minimise side effects. (Grade D, Level 4, 1)
- If an adequate dose of the first drug used demonstrated limited response or was poorly tolerated, change to a different drug class instead of increasing the dose of the first drug. (Grade A, Level 1+, 1)
- Add a second drug when a single drug fails to achieve target BP. (Grade B, Level 1+, 1)
- Use long-acting drugs which provide 24-hour efficacy daily. (Grade B, Level 2+, 1)

Choice of antihypertensive drugs

- In hypertensive patients without compelling indications or contraindications for any particular drug, consider any one, or any appropriate combination, of the four major pharmacological classes of anti-hypertensive drugs (ACE inhibitor /ARB, calcium channel blockers and diuretics).
- Take compelling indications and contraindications into account when prescribing an antihypertensive drug. (Grade A, Level 1+)
- Begin first-line antihypertensive treatment with any one, or an appropriate combination, of the four major drug classes available in Singapore, namely: Angiotensin-converting enzyme inhibitor (ACE inhibitor), Angiotensin II receptor blocker (ARB), Calcium-channel blocker (CCB), Diuretic (thiazide, thiazide-like, or loop).
- Other classes of antihypertensive drugs, such as methyldopa, hydralazine, and alpha-adrenergic receptor blockers may be used in combination treatment as third or fourth-line agents.

Antihypertensives available in NUP	Recommended Dose Range	Renal Dose Adjustment	Common ADR	Contraindications / Precautions
Angiotensin Converting Enzyme Inhibitors (ACE-I)				
Lisinopril (S1) (5 mg, 10 mg and 20 mg tablets) \$^A	5 mg OD (elderly 2.5 mg OD) to 40mg OD	CrCl < 10, initial: 2.5mg OD CrCl 10- 30, Initial 2.5mg – 5mg OM	• Postural hypotension • Dizziness • Abnormal taste • Dry cough • Hyperkalaemia	• Bilateral renal artery stenosis • Pregnancy (2nd / 3rd trimester) • Idiopathic or hereditary angioedema
Enalapril (S1) (5 mg, 10 mg and 20 mg tablets) \$^A	5 mg OM (elderly 2.5 mg OD) to 20mg BD	CrCl 10–30, initial: 2.5 mg/day in 1–2 divided dose; max: 20 mg/day CrCl <10, initial: 1.25mg OD or 2.5mg EOD, max: 10mg/day		
Captopril (S1) (12.5 mg and 25 mg tablets) \$\$-\$\$\$^A	25 mg BD/TDS to 50 mg TDS	CrCl 10–50, administer 75% of normal dose every 12–18h, Max 50mg BD CrCl<10, initial: 1.25mg OD, Max 50mg OD		
Perindopril (NS) (4 mg tablets erbumine) \$\$-\$\$\$^A	4 mg OD (elderly 2 mg OD) to 8 mg OD	CrCl 30–80, initial: 2 mg OM, max: 8 mg OM CrCl < 30, not recommended		
Angiotensin Receptor Blockers (ARB)				
Losartan (S2) (50 mg and 100 mg tablet) \$-\$\$^A	25 mg OD to 100 mg OD	No dose adjustment needed	• Postural hypotension • Dizziness • Fatigue • Hyperkalaemia	• Bilateral Renal artery stenosis • Pregnancy (2nd / 3rd trimester)
Valsartan (S2) (80 mg and 160 mg tablets) \$\$-\$\$\$\$^A	80 mg OD to 320 mg OD	No dose adjustment needed		
Irbesartan (NS) (150 mg and 300 mg tablets) \$\$^A	150 mg OD (elderly 75 mg OD) to 300 mg OD	No dose adjustment needed		
Telmisartan (S2) (40 mg and 80 mg tablets) \$\$^A	40 mg OD to 80 mg OD	No dose adjustment needed		
Candesartan (NS) (8 mg tablets) \$\$-\$\$\$^A	8 mg OD to 32 mg OD	Renal impairment, Initial: 4mg OM CrCl<30, Max 16mg OD		

Antihypertensives available in NUP	Recommended Dose Range	Renal Dose Adjustment	Common ADR	Contraindications / Precautions
Calcium Channel Blockers				
Amlodipine (S1) (5 mg and 10 mg tablets) \$-\$\$\$^A	2.5 mg OD to 10 mg OD	No dose adjustment needed	- Peripheral oedema - Dizziness - Headache / Flushing	Caution use in patients with Heart failure. If to use, amlodipine or felodipine are the preferred choice
Nifedipine LA (S2) (30 mg and 60 mg tablet) \$-\$\$\$^A	30 mg OD to 120 mg OD	No dose adjustment needed		
Diltiazem (S1) Tablets 30 mg and 60 mg \$\$\$-\$\$\$\$\$^A SR Capsules 90 mg, 100 mg and 200 mg \$\$ - \$\$\$\$^A	30 mg TDS to 60 mg TDS	Use with caution in renal impairment	- Peripheral Oedema - Headache	- Sick sinus syndrome - Second- or third-degree AV block - Acute MI - Pulmonary Congestion - Caution use in patients with heart failure
	90 mg OD to 200 mg OD			
Beta-Blockers				
Atenolol (S1) (50 mg and 100 mg tablets) \$^A	25 mg OD to 100 mg OD	CrCl 15–35, max: 50 mg OD CrCl<15, Max 25mg OD or 50mg EOD	- Postural hypotension - Tiredness / fatigue - Cold extremities	- Asthma - Sinus node Dysfunction - Pregnancy - Uncompensated heart failure - Heart block greater than 1st degree
Bisoprolol (S2) (2.5 mg and 5 mg tablets) \$^A	1.25 mg OD to 10 mg OD	CrCl <20, initial: lower initial dose, Max: 10 mg/day		
Carvedilol (S2) (6.25 mg and 25 mg tablets) \$-\$\$\$\$\$^A	6.25 mg BD to 25 mg BD	No dose adjustment needed		
Metoprolol (NS) (50 mg and 100 mg tablets) \$\$-\$\$\$\$\$^A	50 mg BD to 100 mg BD	No dose adjustment needed		

Antihypertensives available in NUP	Recommended Dose Range	Renal Dose Adjustment	Common ADR	Contraindications / Precautions
Thiazide Diuretics				
Hydrochlorothiazide (S1) (25 mg tablet) \$^A	12.5 mg OD to 25 mg OD	Use with caution in renal impairment. CrCl < 10, use not recommended due to lack of efficacy	- Postural hypotension - Electrolytes disturbances (hypokalaemia - more likely for doses ≥ 25mg OD)	- Pregnancy - Renal decompensation - Anuria - May precipitate gout - Increased risk of non-melanotic skin cancer*
Indapamide 2.5 mg (S2) Indapamide SR 1.5 mg (NS) \$-\$\$^A	2.5 mg OD to 5 mg OD SR 1.5 mg OD	CrCl < 30 use not recommended		- Sulphonamides allergy - May precipitate gout - Avoid in severe renal disease (ineffective)
Mineralocorticoid Receptor Antagonist (MRA)				
Spironolactone (S1) (25 mg tablets) \$^A (for resistant hypertension or persistent albuminuria caused by hypertensive kidney disease)	25 mg OD to 50 mg OD/BD	Use with caution in patients with renal impairment. Risk of hyperkalaemia. CrCl < 30 use not recommended	- Breast tenderness or gynaecomastia (~6%), impotence in men, and menstrual irregularities in women - Hyperkalaemia especially when taken together with ACE-I/ARBs	- Should usually be restricted to patients with an eGFR ≥ 45 mL/min and a plasma potassium concentration of ≤ 4.5 mmol/L - Electrolytes and eGFR should be monitored soon after initiation and at least annually thereafter
Vasodilators				
Hydralazine (S1) (10 mg, 25 mg and 50 mg tablets) \$\$^A	10 mg TDS to 50 mg TDS	No dose adjustment needed	- Tachycardia - Flushing - Peripheral oedema	- Mitral valve rheumatic heart disease - May caused a drug-induced Lupus- like syndrome (more likely with larger dose, longer duration)
Alpha-blockers				
Prazosin (S1) (1 mg) \$^A	0.5 mg TDS to 10 mg BD	No dose adjustment needed	- Postural Hypotension - Fatigue	

* [*Risk of Non-Melanoma Skin Cancer Associated with Use of Hydrochlorothiazide*](#)

Antihypertensives available in NUP	Recommended Dose Range	Renal Dose Adjustment	Common ADR	Contraindications / Precautions
Centrally Acting Agent				
Methyldopa (S1) (250 mg Tab) \$^A	250 mg BD/TDS to 500 mg TDS	No dose adjustment needed	Postural Hypotension	- Current MAOI therapy - Acute liver disease
Single-pill combinations				
Hyzaar and Hyzaar Forte (S2) Tablet Hyzaar® - 50 mg / 12.5 mg \$^A Hyzaar Forte® - 100 mg / 25 mg) \$^A	Initial: Hyzaar 1 tab OD Max: Hyzaar Forte 1 tab OD	No dose adjustment needed	- Postural hypotension - Dizziness - Fatigue	- Bilateral Renal artery stenosis - Pregnancy (2nd / 3rd trimester)
Co-diovan (NS) Valsartan 80 mg/HCTZ 12.5 mg \$\$\$\$^A Valsartan 160 mg/HCTZ 12.5 mg \$\$\$\$^A	Initial: I80/12.5 mg OD Max: 160/12.5 mg OD	Use with caution in renal impairment. CrCl < 10, use not recommended due to lack of efficacy		
Co-aprovel (NS) Irbesartan 150 mg/HCTZ 12.5 mg \$\$^A (Not available in NUP)	Initial: 150/12.5 mg OD Max: 300/12.5 mg OD	Use with caution in renal impairment. CrCl < 10, use not recommended due to lack of efficacy		
Nifedipine and Atenolol (NS) Capsule (20/50 mg) \$\$\$^A	Initial: 1 cap OD Max: 1 cap BD	No dose adjustment needed	Similar to calcium channel blockers and beta blockers	
Entresto (NS) Entresto 50 mg - Sacubitril 24 mg/Valsartan 26 mg \$\$\$\$^A Entresto 100 mg - Sacubitril 49 mg/Valsartan 51 mg \$\$\$\$^A Entresto 200 mg - Sacubitril 97 mg/Valsartan 103 mg \$\$\$\$\$^A	Initial: 100 mg OD Max: 400 mg OD		- Postural hypotension - Dizziness - Fatigue - Hyperkalaemia	- Bilateral Renal artery stenosis - Pregnancy (2nd / 3rd trimester) - History of angioedema with ACE inhibitors or ARBs

^ Cost	\$	<\$10 per month	\$\$\$\$	\$30 - <\$60 per month
	\$	\$10 - <\$20 per month	\$\$\$\$\$	\$60 - <\$90 per month
	\$	\$20 - <\$30 per month	\$\$\$\$\$	\$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

Home Blood Pressure Monitoring

1. Indications for Home Blood Pressure Monitoring (HBPM) or 24-Hour Ambulatory Blood Pressure Monitoring (ABPM) include:

- Diagnosis of hypertension, e.g., for cases whose readings in the clinic are repeatedly borderline or show unusual variability.
- When white-coat hypertension (e.g., high clinic blood pressure readings without target organ damage) or masked hypertension (e.g., target organ damage with normal clinic blood pressure readings) is suspected.
- Monitoring of blood pressure in treated hypertensive patients.
- Symptoms suggesting hypotension from any likely cause, such as postural, autonomic, postprandial, afternoon nap-related and drug-induced blood pressure fall.
- Elevated clinic blood pressure or suspected pre-eclampsia in pregnant women.
- Identification of true and false resistant hypertension.
- Specific indications for 24-hour ABPM:
 - Extreme discordance between clinic and home blood pressure readings.
 - Assessment of intra-day blood pressure variability.
 - Evaluation of nocturnal blood pressure dipping status.
 - Suspicion of nocturnal hypertension, or absence of night-dipping (e.g., patients with diabetes, chronic kidney disease, obstructive sleep apnoea and long-term night-shift workers).

2. Advantages of HBPM

- Multiple measurements of BP and HR can be made during the day and also over several days, weeks or months
- Assessment of treatment effects at different times of the day and over extended periods is possible
- Good reproducibility, good prognostic value, relatively low cost (compared to ABPM)
- Improvement in patient engagement with long-term treatment
- May empower patients in BP management, including adherence to therapy

3. Requirements of HBPM

- Patient education on HBPM technique is required
- Home BP monitor should be validated yearly
- To note that cardiac arrhythmias (particularly AF and bradycardia) may affect accuracy of BP measurement; manual BP should be checked at clinic visits

4. Advice to be given to patients placed on HBPM

- Preparation and positioning to follow office BP measurement using electronic BP machine (see above)
- To avoid talking or movement during BP measurement
- To record results manually or store in the device memory immediately after measurement

5. Monitoring Schedule

- For diagnosis of hypertension, monitoring of those with high normal BP, and assessment of the effects of antihypertensive treatment (including changes in drug or dose)
 - Seven-day home measurements (minimum 3 days) before each clinic / tele visit
 - Two readings per day: Morning (before medication intake if on treatment) and evening (before eating)
 - Two measurements each time (1-2 minutes apart)
- Long-term follow-up
 - Less frequent measurements (e.g., once or twice per week) are acceptable
 - Overuse of HBPM and self-modification of treatment should be avoided

6. Interpretation of HBPM Readings

- Compute average of HBPM readings but exclude readings from the first day (i.e., start date of HBPM)
- Mean home systolic BP ≥ 135 mmHg and/or diastolic BP ≥ 85 mmHg indicates hypertension
- Mean home systolic BP < 130 mmHg and diastolic BP < 80 mmHg should be considered normal
- Mean systolic BP of 130-134 and diastolic BP of 80-84 requires continued regular monitoring (watch for hypertension)
- Diagnosis thresholds and treatment targets for HBPM readings are generally 5 mmHg lower (for systolic and diastolic) compared to office BP measurements. For example, home BP threshold for diagnosis of hypertension is $\geq 135/85$ mmHg, and home BP target for patients with diabetes is $< 135/75$ mmHg (see [Initial Screening Blood Pressure](#) and [Treatment Targets](#)).

Role of Health Team Members

1. Family Physician

- Able to provide all aspects of primary medical care from screening, diagnosis and management of hypertension, including health promotion and prevention / treatment of complications
- Collaborates with other health care providers in the health team to provide holistic and patient-centred care

2. Care Coordinator

- Introduce OneNUHS and HealthHub Apps to patient
- Introduce and enrol patient onto Teleconsult
- Perform general screening (i.e., fall risk, social economics, smoking & drinking history)
- Address care gaps under Health Maintenance Topics, such as vaccinations (influenza / pneumococcal) and cancer screenings (FIT kit, mammogram, PAP smear / HPV)
- Teach home blood pressure monitoring to patient
- Recruitment into PTEC - HT (Inclusion criteria: Any diagnosed hypertensive patient. Exclusion criteria: Atrial fibrillation / arrhythmias, CKD G3b - G5, Pregnant women and unstable chronic conditions)

3. Care Manager

- Patient education on hypertension, BP target and lifestyle measures
- Teach home BP monitoring and validate BP set
- Assess for treatment adherence
- Alternate CM visit with Doctor for patients with stable hypertension and/or dyslipidaemia
- Recruitment into PTEC - HT

4. Advanced Practice Nurse

- Manage patients with diabetes, hypertension and dyslipidaemia within scope of practice
- Initiate and titrate antihypertensive medications
- Collaborate with other healthcare providers in the health care team to provide holistic care.

5. Clinical Pharmacist

- Manage patients with diabetes, hypertension and dyslipidaemia
- Initiate and titrate antihypertensive medications
- For patients who need drug optimisation or closer monitoring due to increased risk for drug interactions, polypharmacy, lack of drug-related knowledge or drug non-adherence

6. Dietitian

- Patient education on DASH diet
- Patient education on weight management
- Patient who needed more dietary / nutritional support / advise on food choices

7. Psychologist

- Psychological and behavioural interventions to manage psychological stress, improve disease management and quality of life.
- Psychological assessment and intervention to manage co-occurring psychological problems (depression, anxiety disorders etc.)
- Psychological support for carers experiencing caregivers stress

8. Medical Social Worker

- Conduct biopsychosocial assessment
- Provide intervention for social assistance including application of schemes and services, and referral for community resources and caregiver support to support with living with their chronic disease

9. Financial Counsellor

- Financial counselling advice with patients raising difficulties in managing their medical bills
- Advise on MediSave utilisation for chronic disease polyclinic bills
- Application on government endowment fund- MediFund and Medication Assistance Fund for subsidies on high-cost drugs
- Application on Institutional Medical Fund-Remission (if applicable for non-Singaporeans)

10. Pharmacist

- Smoking cessation clinic
- Detect, prevent and address drug-drug / drug-disease interactions
- Perform medication reconciliation.

SPECIAL SITUATIONS

Microscopic Haematuria or Micro-/Macro-albuminuria

What to do if a hypertensive patient is found to have microscopic haematuria or micro- / macro-albuminuria?

- Please refer to the guideline on chronic kidney disease.

Secondary Hypertension

- Secondary hypertension is hypertension due to an identifiable cause, which may be treatable with an intervention specific to the cause.
- A high index of suspicion and early detection of secondary causes of hypertension are important because interventions may be curative, especially in younger patients.
- Patient characteristics that should raise the suspicion of secondary hypertension:
 - Younger patients (< 40 years) with grade 2 hypertension or onset of any grade of hypertension in childhood
 - Acute worsening hypertension in patients with previously documented chronically stable normotension
 - Resistant hypertension
 - Severe (grade 3) hypertension or a hypertension emergency
 - Presence of extensive hypertension-mediated organ damage (HMOD)
 - Clinical or biochemical features suggestive of endocrine causes of hypertension or CKD
 - Clinical features suggestive of obstructive sleep apnoea
 - Symptoms suggestive of pheochromocytoma or family history of pheochromocytoma

Causes:

- Obstructive sleep apnoea (5 - 10%)
- Renal parenchyma disease (2 - 10%)
- Primary hyperaldosteronism (5 - 15%)
- Atherosclerotic renal vascular disease (1 - 10%)
- Thyroid disease (1 - 2%)
- Pheochromocytoma, Cushing's syndrome, coarctation of the aorta, hyperparathyroidism (< 1%)
- Medications and other substances that may increase blood pressure:
 - Oral contraceptive pill (especially oestrogen containing pills)
 - Diet pills (e.g., phenylpropanolamine and sibutramine)
 - Nasal decongestants

- Stimulant drugs (amphetamine, cocaine, ecstasy – cause acute rather than chronic hypertension)
- Immunosuppressive medications (e.g., cyclosporin A and steroids)
- Anti-angiogenic cancer therapies (e.g., VEGF inhibitors (e.g., bevacizumab), tyrosine kinase inhibitors and sorafenib)
- Other (anabolic steroids, erythropoietin, NSAIDs, and herbal remedies like ephedra and *ma huang*)

Hypertensive Emergencies and Urgencies

1. Hypertensive Emergencies

- Large elevations in blood pressure (systolic blood pressure > 180 mmHg or diastolic blood pressure > 110 mmHg) associated with impending or progressive organ damage, e.g., major neurological changes, hypertensive encephalopathy, cerebral infarction, intracranial haemorrhage, acute LV failure, acute pulmonary oedema, aortic dissection, renal failure, eclampsia.
- Referral to the Emergency Department is indicated.

2. Hypertensive Urgencies

- Isolated large blood pressure elevations without acute target organ damage.
- Often associated with treatment discontinuation / reduction as well as anxiety.
- Exclude acute target organ damage through history (symptoms), physical examination including fundoscopy, urinalysis (for proteinuria / haematuria) ± serum creatinine.
- Can be treated by reinstitution or intensification of drug therapy and treatment of anxiety.
- BP may not settle within 1 to 2 hours, both due to pharmacokinetics / onset of action of drugs as well as patients' anxiety / impatience while waiting in the clinic.
- Once target organ damage has been excluded, consider sending home (with home BP monitoring if available), with scheduled review on the following day or within the next few days.
- Patient to proceed to Emergency Department if symptomatic.

Resistant Hypertension

- Blood pressure that remains above goal (average blood pressure > 140/90 mmHg) despite concurrent use of three antihypertensive agents of different classes at optimal doses (or best tolerated doses), one of which should be a diuretic.
- The inadequate control of BP is confirmed by ABPM or HBPM in patients whose adherence to therapy has been confirmed.
- Resistant hypertension is associated with older age (especially > 75 years), male sex, black African origin, higher initial BP at diagnosis of hypertension, highest BP ever reached during the patient's lifetime, frequent outpatient visits, obesity, diabetes, atherosclerotic disease and hypertension-mediated organ damage, CKD, and a Framingham 10-year coronary risk score > 20%
- Evaluation
 - Detailed history / exam / investigations
 - Exclude secondary cause

Spurious Causes	True Causes
Non-adherence to medication	Lifestyle factors (obesity / large weight gain, excessive alcohol consumption, high sodium intake)
"White-coat" effect	Chronic intake of vasopressor / sodium-retaining substances (sympathomimetics, nasal decongestants, oral contraceptives, non-steroidal anti-inflammatory drugs)
Wrong cuff size	Obstructive sleep apnoea, chronic pain
Pseudo-hypertension - Marked arterial stiffening (elderly with calcified arteries) preventing occlusion of brachial artery	Secondary hypertension
	Advanced end-organ damage, e.g., renal impairment

- Possible measures
 - Non-pharmacological
 - Check and reinforce adherence to medication and diet / lifestyle measures (especially reduction of sodium intake).
 - Address any drug interactions and associated medical problems, if present.
 - Pharmacological
 - Addition of low-dose spironolactone to existing treatment
 - Or the addition of further diuretic therapy if intolerant to spironolactone, with either eplerenone, amiloride (not available in NUP), a higher dose thiazide/thiazide-like diuretic, or a loop diuretic for patients with renal impairment (eGFR $\leq$ 30 mL/min);
 - Or the addition of bisoprolol.

White Coat Hypertension

- Suggested by:
 - Markedly elevated clinic blood pressure readings in the absence of documented end-organ damage.
 - Normal ambulatory blood pressure readings taken at work or at home.
 - Unusual variability of blood pressure readings.
 - Symptoms suggesting episodes of hypotension.
 - Blood pressure seemingly resistant to treatment.
- Ambulatory or home blood pressure monitoring useful for identification and monitoring.

REFERRALS

Referral to Hospital

1. Patients in whom secondary hypertension is suspected
2. Younger patients (< 40 years) with grade 2 or more severe hypertension in whom secondary hypertension should be excluded
3. Patients with treatment-resistant hypertension
4. Patients in whom more detailed assessment of HMOD would influence treatment decisions
5. Patients with sudden onset of hypertension when BP has previously been normal
6. Other clinical circumstances in which the referring doctor feels more specialist evaluation is required
7. When there is a need for 24-hour ambulatory BP monitoring
8. Hypertensive emergencies and urgencies (when it is not feasible to treat and monitor in the clinic) – refer to hospital A&E

RECOMMENDED CARE COMPONENTS FOR HYPERTENSION

Recommended Care Components	Minimum Frequency	Remarks
BP measurement~	Twice a year	
Weight and BMI measurement~	Twice a year	Keep < 23kg/m ² (For non-Asian population, keep BMI < 25 kg/m ²)
Kidney Assessment~ (i) Serum Cr and/or eGFR, and (ii) Urine Albumin-Creatinine Ratio (uACR) or Protein-Creatinine Ratio (uPCR)	Annually	Annual screening of (i) Serum Cr and/or eGFR and (ii) uACR in all patients, or uPCR if significant levels of proteinuria
Smoking assessment~	Annually for smokers; Once-off for non-smokers, unless there is a change in smoking habit	Assessment on smoking habits (estimated sticks/day; zero for non- or ex-smoker) and provide smoking cessation counselling
Lipid profile	At or soon after diagnosis	All patients should be risk stratified (as recommended in the Lipids CPG) Targets of treatment should be personalised by levels of risk
Cardiac assessment	At diagnosis before initiating medications	Includes baseline ECG

~ These are reportable clinical indicators

ABBREVIATIONS

ABPM	Ambulatory blood pressure monitoring
ACE-I	Angiotensin-converting enzyme inhibitor
ARB	Angiotensin receptor blocker
BMI	Body mass index
BP	Blood pressure
CCB	Calcium channel blocker
CKD	Chronic kidney disease
DBP	Diastolic blood pressure
HBPM	Home blood pressure monitoring
HMOD	Hypertension-Mediated Organ Damage
HR	Heart rate
IHD	Ischaemic heart disease
LM	Lifestyle modification
LVH	Left ventricular hypertrophy
Rx	Pharmacological treatment
SBP	Systolic blood pressure
TIA	Transient ischaemic attack
TOD	Target organ damage
uACR	Urine albumin-creatinine ratio
uPCR	Urine protein-creatinine ratio

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